

Figure 1: The MozStumbler app

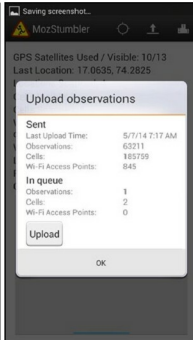


Figure 2: MozStumbler options

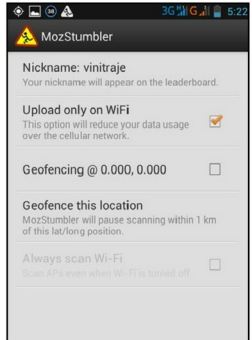


Figure 3: MozStumbler settings

Mozilla's location service website, "The Mozilla Location Service is a research project to investigate crowd-sourced mapping of wireless networks (Wi-Fi access points, cell towers, etc) around the world. Mobile devices and desktop computers commonly use this information to figure out their location when GPS satellites are not accessible."

In the same statement, Mozilla acknowledges the presence of and the challenges presented by the other services, saying, "There are few high-quality sources for this kind of geolocation data currently open to the public. The Mozilla Location Service aims to address this issue by providing an open service to provide location data."

This service provides geolocation lookups based on publicly observable cell tower and Wi-Fi access point information. Mozilla has come out with an Android app to collect publicly observable cell towers and Wi-Fi data; it's called MozStumbler.

This app scans and uploads information of cell towers and Wi-Fi access points to Mozilla servers. The latest stable version of this app is ver 0.20.5 which is ready for download. MozStumbler provides the option to upload this scanned data over a Wi-Fi or cellular network. But you don't need to be online while scanning; you can upload this data afterwards.



Note: 1. This app is not available on Google Play store but you can download it from <https://github.com/MozStumbler/releases/>
2. The Firefox OS version of this app is on its way too. You can stay abreast of what's happening with the Firefox OS app at <http://github.com/FxStumbler/>

You can optionally give your username in this app to track your contributions. Mozilla has also created a leader board to let users track and rank their contributions, apart from more detailed statistics that are available on this website. No user identifiable information is collected through this app.

Mozilla is not only collecting the data but also providing users with a publicly accessible API. It has code named the API 'Ichnaea', which means 'the tracker'. This API can be accessed to submit data, search data or search your location. As the data collection is still in progress, it is not recommended to use this service for commercial applications, but you can try it out on your own just for fun.



Note: Mozilla Ichnaea can be accessed at <https://mozilla-ichnaea.readthedocs.org>

The MozStumbler app provides an option for geofencing, which means you can pause the scanning within a one km radius of the desired location. This deals with user concerns over collecting behavioural commute data such as *Home*, *Work* and travelling habits.

In short, Mozilla is trying to provide a high quality location service to the general public at no cost! Recently, Mozilla India held a competition 'Mozilla Geolocation Pilot Project India', which encouraged more and more users to scan their area. To contribute to this project, you can fork the repository on *github* or just install the app; you will be welcomed aboard.

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The author is a fan of free and open source software. He is currently contributing to the translation of the MozStumbler app for Mozilla location services.